

Micro CPS study – Data cleaning and construction of analysis sample

1. Research question (short reminder)

We study how individual characteristics – especially education – are related to the probability of being unemployed in the United States, using CPS micro data (Basic Monthly, March 2023). The unit of observation is the individual, and the outcome is an employment/unemployment indicator.

2. Raw CPS data and initial variables

Source:

- IPUMS CPS, Basic Monthly sample for March 2023.
- The raw extract contains standard CPS variables such as YEAR, MONTH, AGE, SEX, RACE, EDUC, EMPSTAT and the person weight WTFINL.

Key raw variables:

- YEAR: survey year (2023 for all observations).
- MONTH: survey month (March for all observations).
- AGE: age in completed years.
- SEX: sex of the respondent (1 = male, 2 = female).
- RACE: detailed race category (100 = White, 200 = Black, etc.).
- EDUC: highest educational attainment (harmonised IPUMS coding).
- EMPSTAT: detailed labour-force status (e.g., 10 = at work, 12 = has job, 20–22 = unemployed).
- WTFINL: final CPS person-level sampling weight.

3. Construction of the analysis sample (detailed cleaning steps)

(1) Restrict to working-age individuals

We keep only individuals with AGE ≥ 16 . This is a standard convention to focus on the working-age population. Children and very young teenagers are not considered potential labour-force participants.

(2) Restrict to the labour force

We keep only individuals whose EMPSTAT indicates they are in the labour force:

- employed: EMPSTAT in {10, 12} (at work or has a job but not at work);
- unemployed: EMPSTAT in {20, 21, 22} (different CPS unemployment categories).

All other values of EMPSTAT (students out of the labour force, retirees, disabled, etc.) are dropped. This is essential because unemployment is defined relative to the labour force, not the entire population.

(3) Construct a binary employment status

Rather than using EMPSTAT directly in the analysis, we build a simple 0/1 variable:

- EMPLOYED = 1 if EMPSTAT in {10, 12};
- EMPLOYED = 0 if EMPSTAT in {20, 21, 22}.

This step makes the dependent variable explicit and transparent. It also ensures that our definition of employment/unemployment is perfectly aligned with CPS codes. Any pre-existing variable in the raw file is ignored; we rely entirely on EMPSTAT to avoid inconsistencies.

(4) Create simple profile dummies

To keep the empirical model easy to read and to avoid too many categories, we recode several raw variables into binary indicators:

(a) Gender dummy

- FEMALE = 1 if SEX == 2 (female), 0 otherwise (male).

(b) Education dummy

We want a simple distinction between “high education” and “low/medium education”. Using the IPUMS coding of EDUC, we define:

- HIGHED = 1 if EDUC >= 100, corresponding roughly to college / bachelor or more;
- HIGHED = 0 otherwise (no tertiary degree).

This allows us to ask: is the probability of employment significantly higher for individuals with tertiary education?

(c) Age profile dummy

To capture the difference between younger and older workers without imposing a specific functional form of age, we define:

- YOUNG = 1 if AGE < 35 (16–34 years old),
- YOUNG = 0 otherwise (35 years and older).

This very simple split is enough to highlight that unemployment tends to be higher among young workers.

(d) Minority status dummy

The raw RACE variable is very detailed. For our baseline analysis, we reduce this to a majority/minority split:

- MINORITY = 0 if RACE == 100 (White, considered as the majority group),
- MINORITY = 1 otherwise (all non-White categories).

This parsimonious definition allows us to document whether labour-market outcomes differ between the majority group and minorities, without introducing many small categories that would make interpretation harder.

(5) Handling weights

We keep WTFINL as the unique person-level sampling weight. All descriptive statistics and econometric estimations will be computed using this weight so that our results are representative of the CPS target population. Observations with missing weights (if any) are dropped.

(6) Final cleaned dataset

After all these transformations, we construct a compact analysis dataset `df_clean` with the following variables:

- YEAR : survey year.
- AGE : age in years.
- EMPSTAT : original detailed employment status.
- EMPLOYED : 1 if employed (EMPSTAT 10 or 12), 0 if unemployed (EMPSTAT 20–22).
- SEX : original sex code.
- FEMALE : 1 if female, 0 if male.
- RACE : original race code.
- MINORITY : 1 for non-White groups, 0 for White.
- EDUC : original education code.
- HIGHED : 1 for tertiary education, 0 otherwise.
- YOUNG : 1 if 16–34 years old, 0 if 35+.
- WTFINL : CPS sampling weight.

This cleaned dataset is the only one used for the subsequent descriptive statistics and

econometric analysis. Keeping both raw and recoded variables in `df_clean` gives us flexibility: we can easily refine definitions (e.g. change the `HIGHED` cut-off) without going back to the original extract.

4. Brief descriptive snapshot (no detailed comments)

Using the cleaned CPS sample and `WTFINL` weights, we compute basic descriptive statistics:

- The overall employment rate is about 96.3%, implying an unemployment rate of roughly 3.7% among the labour force.
- Employment rates differ across groups (high vs low/medium education, majority vs minority, young vs 35+), and the shares of each group in the sample (e.g. share of highly educated, minorities, women, young) are documented in a descriptive table.

These numbers simply provide context and confirm that unemployment is relatively low overall but more concentrated among some groups. The main focus of the project is the econometric analysis of the employment/unemployment probability at the micro level.